Course program and reading list

Semester 1 Year 2026

School: Adelson School of Entrepreneurship B.A

Computational Thinking and Programming in Python

Lecturers:

Mr. Sahar Millis sahar.millis@post.runi.ac.il

Tutors:

Mr. Ben Sapirstein ben.sapirstein@post.runi.ac.il

Teaching Assistants:

Ms. Shira Borochovich shira.borochovich@post.runi.ac.il

Course No.:	Course Type :	Weekly Hours :	Credit:
26027	Lecture	4	4

Course Requirements :	Group Code :	Language:
Final Exam	261260270	Hebrew

Prerequisites

Students who took one of the courses listed below will not be allowed to register to the course Computational Thinking and Programming in Python (26027):

26002 - Computational Thinking and Programming

Course Description

This course equips you with practical computational thinking to tackle challenges in business, economics, and entrepreneurship. You'll develop foundational skills in software development, programming, and data analysis - utilizing Python and AI tools. Whether you're new to programming or have some experience, you'll graduate with a powerful skill set to apply in your studies or entrepreneurial pursuits.



Course Goals

Unit #	Title	Topics	Practical Application	Assignment	Weekly Exam
1	Introductions	Computer Science, Python, AI, Computational Thinking	Welcome to Python & Computer Science		
2	Python Fundamentals I	Variables, Data Types	Exploring Variables and Data Types	#1	
3	Python Fundamentals II	Conditional Statements, Loops	Mastering Conditional Statements & Loops	#2	
4	Data Structures	Lists, Dicts, Sets, Objects	Text to Speech	Exploring Data Structure	#3
5	Modular Programming	Functions, Libraries	Manipulation of Data Structure		
6	OOP	Classes	ChatBot	Exploring Functions and Libraries with Games	#4
7	Error Handling & Modularity	Exceptions, Files, JSON, Modules, PIP	Stock Tracker	#5	
8	Data Manipulation	Pandas, Matplotlib, Kaggle	Algorithms & Visualizations	#6	

9	Algorithms	Searching, Sorting		
10	Advanced Concepts (Optional)	(depends on class progress)	TLV Weather System	#7
11	Web & APIs	Internet basics, APIs, Automation	Google Search	Harnessing Web Search, Web Crawling, and AI
12	AI & No-Code Tools	GenAI, HuggingFace, Prompt Engineering, Low/No-Code tools	User Interface	#8
13	Final Lecture	Recap, GitHub, Agile, Jira, Industry insights, Feedback		
14	Exam Preparation	(Optional)		

- Some adjustments may occur.
- Topics might be adapted based on the class's progress.
- Exact dates for guest lectures will be announced during the course.

Grading

Component	**Percentage**	**Details**
Assignments	8%	Extra Credit, Individual
Weekly Exams	30%	Top 6 of 8
Final Exam	70%	

Learning Outcomes

By the end of the course, you will:

- **Understand core CS concepts**: Code, Variables, Functions, Algorithms, and Data Structures.
- **Write functional Python code**: solve real-world problems using code and a User Interface.
- **Use modern tools**: APIs, no-code AI platforms, Kaggle, and HuggingFace.
- **Build Applications**: such as chatbots, search, and automations.



AI Policy in the Course

Allowing free use of AI, according to the university's policy

The use of Generative Artificial Intelligence (GAI) tools can offer significant educational value. However, it is important to set clear rules and expectations for their use in assignments and exams.
When using GAI tools in assignments or exams, students must report such use within the submission, specifying how the tools were used — for example, the type of AI tool used or the content of the prompts.
The use of GAI tools in violation of these university guidelines, may be considered a disciplinary offense.



Lecturer Office Hours

Before and after class.
Additionally, upon request - online during the week.



Tutor Office Hours

Online during the week (upon request).



Teaching Assistant

TBD



Additional Notes

Course will be handled on Moodle.

Recordings of lessons will be uploaded.

Attendance is aligned with the Entrepreneurship School's guidelines.

About the lecturer.

About the instructor.

Reading List

Online:

1. Python.org
2. W3Schools
2. GeeksforGeeks.org
4. Kaggle.com (Learn)
5. HuggingFace.com (Datasets, Models, Spaces)

Online Course:

1. CS50's Introduction to Computer Science by Harvard
2. Introduction to CS and Programming using Python by OpenMIT
3. Introduction to Computer Science and Programming in Python by MIT

Books (based on lecture order):

1. Allen B. Downey, O'Reilly, 2012, "Think Python".
2. Michael Sipser, 2012, "Introduction to the Theory of Computation".
3. Brian Christian, Griffiths, 2016, "Algorithms to Live By: The Computer Science of Human Decisions".
4. Guttag John, MIT, 2016, "Introduction to Computation and Programming Using Python with Application to Understanding
Data".
5. David Bernstein, 2022, "Patterns for Beginning Programmers".
6.Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, MIT, 2022, "Introduction to Algorithms" (fourth edition).
7. Harris Wan, 2023, "Introduction to Computer Programming with Python".

Generative Tools:

1. groq.com
2. claude.ai

3. chatgpt.ai
4. perplexity.ai
5. gemini.google.com

Coding Frameworks (by complexity):

1. https://colab.google/
2. https://code.visualstudio.com/
3. https://www.jetbrains.com/pycharm/